

3D-Consistent Segmentation and Neural Reconstruction for Plant Timelapses

Juliane Mercoli¹, Kevin Zihlmann¹ Joaquin Gajardo², Dr. Sergey Prokudin³

¹ETH Zurich, D-INFK ²Department of Environmental Systems Science, ETH Zurich ³ETH AI Center, ETH Zurich

1. Problem Statement

Accurate 3D reconstruction of growing plants can support phenotyping, growth monitoring, and automated analysis of plant structure over time. However, plant timelapse reconstruction is difficult because plants are thin, self-occluding, visually repetitive, and occupy only a small fraction of each image. In the greenhouse setup, background objects such as the pot, checkerboard, table, and camera rig can dominate full-image metrics even when the plant itself is poorly reconstructed.

This project plant masks which allows the different reconstruction methods like 3DGS, Nerfacto, and 4DGS to focus on the plant foreground and produce cleaner renderings, silhouettes, and point-cloud outputs.

2. Related Work

We build on 3D Gaussian Splatting for explicit differentiable rendering (Kerbl et al., 2023), 4D Gaussian Splatting for dynamic scenes (Wu et al., 2024), Nerfstudio for NeRF-style reconstruction and point-cloud export (Tancik et al., 2023), and recent agricultural 2D-to-3D reconstruction work motivating semantic foreground guidance (Sankaramaddi et al., 2026).

3. Data

The dataset contains synchronized multi-view timelapse recordings of maize and sunflower plants in a controlled greenhouse. We focus on the maize sequence. Ten equally spaced timesteps, sampled every 30 hours, are used for static 3DGS, Nerfacto, and 4DGS. Each timestep has 22 calibrated GoPro views.

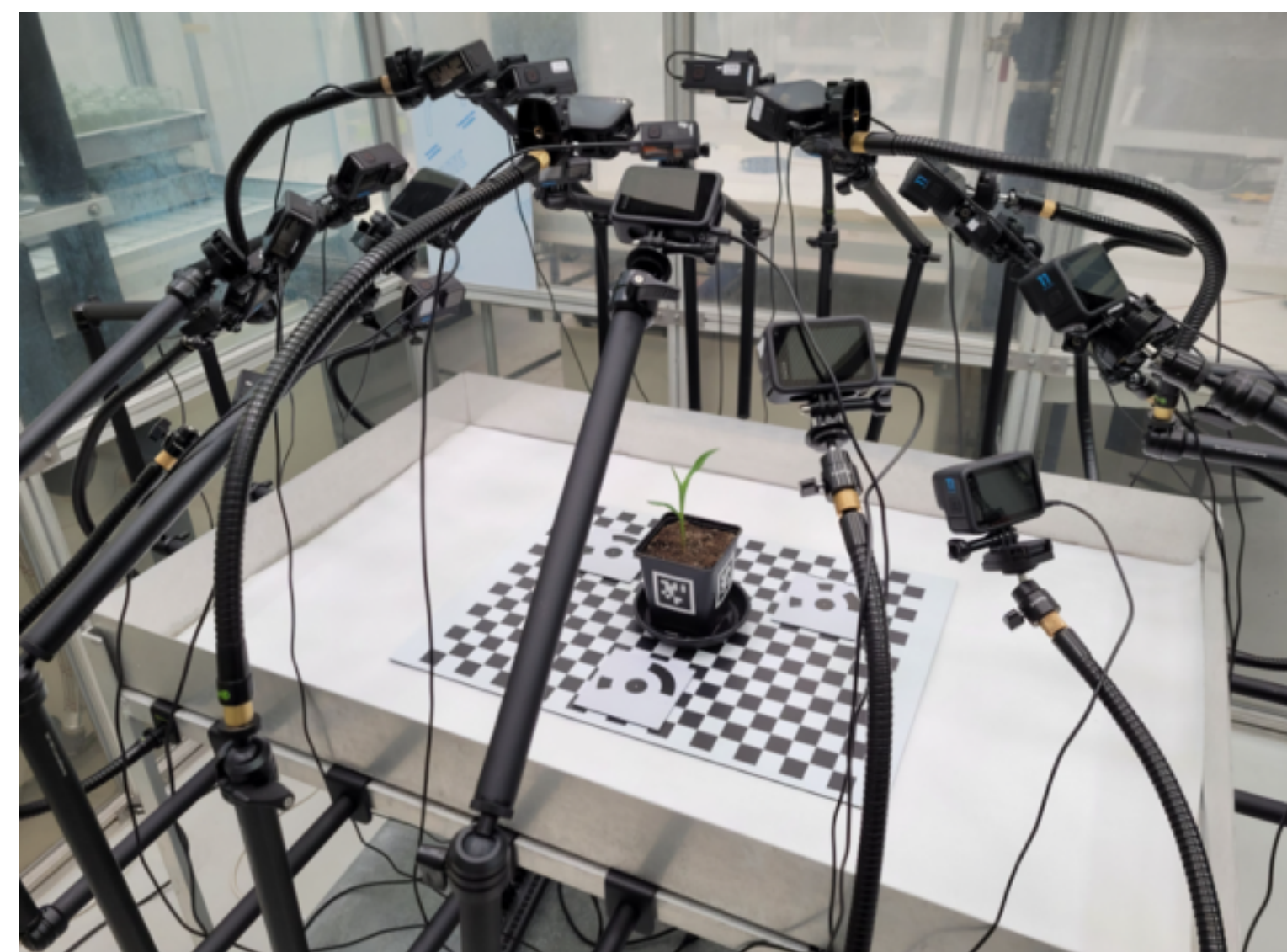


Figure 1: Multi-view greenhouse setup with 22 calibrated GoPro cameras.

4. Methods

- **Segmentation:** plant-background masks are produced using a color baseline, Grounding DINO + SAM, and SAM 3.
- **Static 3DGS:** Wheat3DGS is adapted to plant-only reconstruction using hard-masked RGB images or alpha-mask supervision. The final setup uses foreground-masked photometric loss with opacity-based foreground supervision.
- **Nerfacto RGBA:** SAM 3 masks are inserted as alpha channels in RGBA images. One Nerfacto model is trained per timestep and evaluated with plant-only metrics, RGB-derived silhouettes, and exported PLY point clouds.
- **Dynamic 4DGS:** one shared temporal Gaussian model is trained on ten timesteps using SAM 3 foreground masks, opacity-mask supervision, and foreground-weighted photometric loss.

Evaluation uses plant-only PSNR, silhouette IoU/Dice, and qualitative point-cloud inspection instead of only full-image metrics.

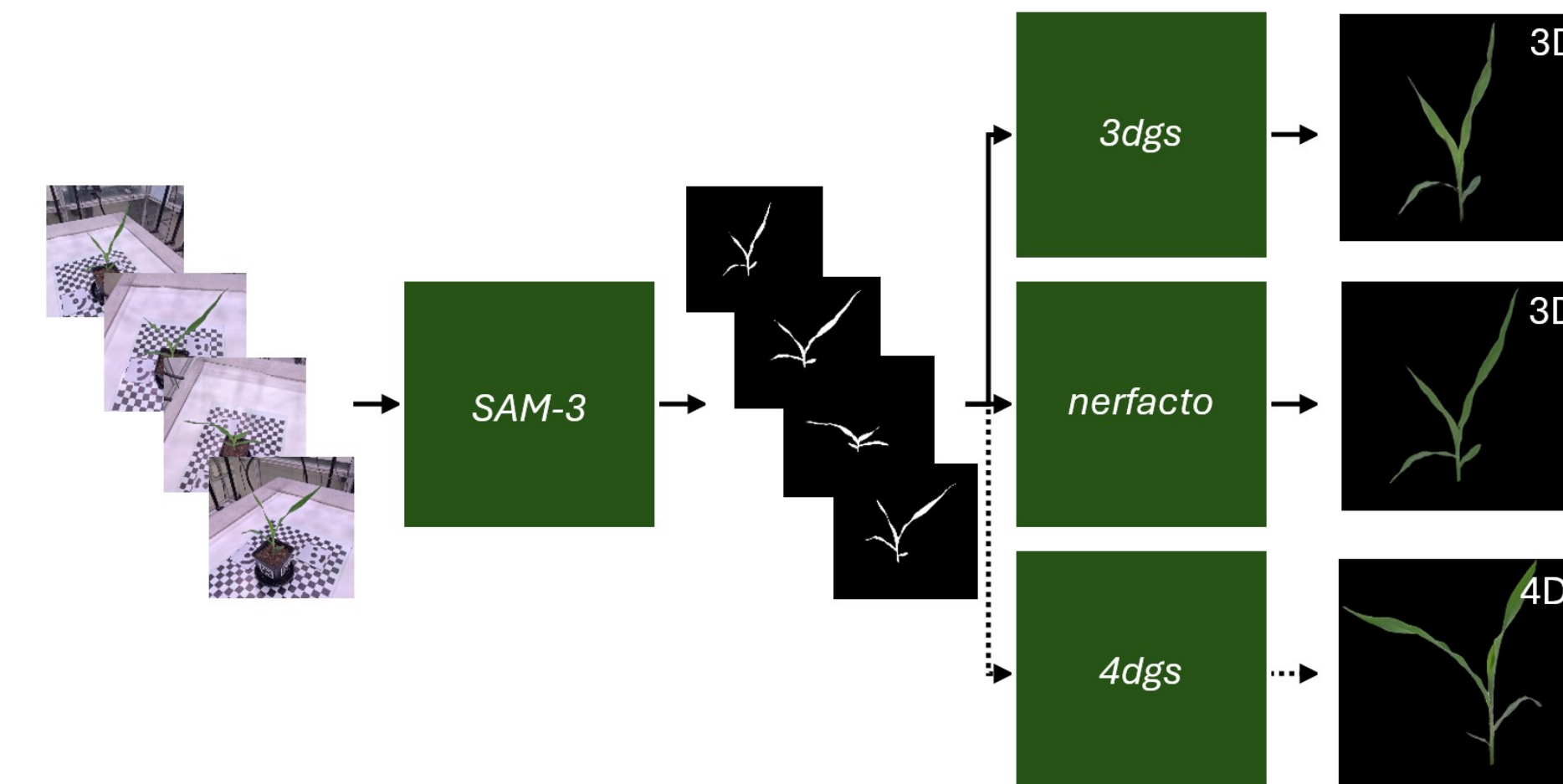


Figure 2: Segmentation-guided reconstruction pipeline.

5. Results

SAM 3 gives the best segmentation performance, reaching mAP 0.975 ± 0.007 and IoU 0.951 ± 0.014 on the annotated validation frames.

For reconstruction, the best method depends on the target output. Static 3DGS gives strong foreground alignment, dynamic 4DGS achieves the highest plant-only PSNR but less precise silhouettes, and Nerfacto RGBA gives the smoothest qualitative point-cloud exports.

Ablations showed that opacity-based alpha supervision was the most useful 3DGS modification. Nerfacto-derived point clouds were useful for visualization, but not as direct 3DGS initializers: Nerfacto-only initialization collapsed, while COLMAP+Nerfacto gave only marginal gains over COLMAP-only 3DGS.

Method	Plant PSNR	Sil. IoU	Sil. Dice
Static 3DGS	17.79 ± 0.28	0.783 ± 0.014	0.877 ± 0.009
Dynamic 4DGS	22.21 ± 2.20	0.640 ± 0.030	0.779 ± 0.022
Nerfacto RGBA	17.20 ± 1.84	0.803 ± 0.052	0.887 ± 0.034

Table 1: Foreground-aware reconstruction comparison across the three reconstruction baselines, averaged over 10 timesteps. Values are reported as mean $\pm$ 90% confidence interval.

Full-image metrics were found to be misleading: Nerfacto reaches full-image PSNR 31.63, but the foreground occupies only 1.07% of pixels on average. Foreground-aware metrics are therefore essential. Raw 3D exports still require plant-centered post-processing: Gaussian splats and point-cloud exports can contain off-plant outliers and elongated artifacts, so cropping/filtering around the plant is needed.



Figure 3: 3D reconstructions using 3DGS (top) and nerfacto (bottom).

6. Conclusions

Segmentation-guided reconstruction reduces the influence of greenhouse background and enables plant-centered evaluation. Static 3DGS is best for explicit foreground-aligned Gaussian reconstruction and mask-consistent silhouettes. Nerfacto RGBA is strongest for smooth qualitative rendering and point-cloud visualization. Dynamic 4DGS improves plant-only PSNR, but still loses boundary precision.

The final pipeline should support both static 3DGS and Nerfacto RGBA: 3DGS for foreground-aware Gaussian reconstruction, and Nerfacto for smooth point-cloud extraction.

7. Future Work

- Automate plant-centered point-cloud cleaning.
- Compare 3DGS and Nerfacto exports with geometric metrics such as compactness, outlier ratio, and temporal smoothness.
- Improve 4DGS with mask-aware densification and temporal priors for sharper leaf boundaries.
- Explore GARfield for part-level 3D segmentation, e.g. separating leaves or coherent leaf clusters instead of only the whole plant foreground.
- Track plant parts across timesteps to analyze leaf growth, movement, and occlusion in a temporally consistent 4D representation.

References

- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023. doi: 10.1145/3592433. URL <https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/>.
- Hemanth Reddy Sankaramaddi, Won Suk Lee, Kyoungchul Kim, and Youngki Hong. 2d-to-3d image reconstruction in agriculture: A review of methods, challenges, and ai-driven opportunities. *Sensors*, 26(6):1775, 2026. doi: 10.3390/s26061775. URL <https://www.mdpi.com/1424-8220/26/6/1775>.
- Matthew Tancik, Ethan Weber, Evonne Ng, Ruilong Li, Brent Yi, Justin Kerr, Terrance Wang, Alexander Kristoffersen, Jake Austin, Kamyar Salahi, Abhik Ahuja, David McAllister, and Angjoo Kanazawa. Nerfstudio: A modular framework for neural radiance field development. In *ACM SIGGRAPH 2023 Conference Proceedings*, 2023.
- Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 4d gaussian splatting for real-time dynamic scene rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20310–20320, 2024. URL <https://arxiv.org/abs/2310.08528>.

Contributions

Author contributions using the CRediT framework (??):
S1: Methodology, Software, Visualization, Investigation, Writing - Original Draft
S2: Methodology, Software, Visualization, Investigation, Writing - Original Draft
CG: Conceptualization, Supervision
AC: Supervision, Methodology, Project Administration

Preprint. This course project work can be distributed as a preprint and has not been peer-reviewed. It does not constitute archival publication and remains eligible for submission to academic venues, including workshops, conferences, and journals.

License. The authors grant ETH Zurich and the ETH AI Center a non-exclusive license to display this work on their platforms to showcase student projects. Redistribution or publication by others outside of academic venues requires the consent of the authors.

Code. Associated code is available open-source and under the MIT License, which permits free reuse, free modification, and free distribution, provided proper attribution is given to the authors, and no liability is assumed by the authors. Follow up work is not required to be open-source and is not required to have an MIT license. The code and its MIT license are publicly available here: <https://github.com/kezihlmann/dsl-p6-pipeline>